

RAMCO Institute of Technology, Rajapalayam

Department of Mechanical Engineering

Academic Year 2020 – 2021 (ODD Semester)

Innovative Practice(s) Description

Name of the Faculty : Mr. J.Jerold John Britto AP(Sr.Gr.)/Mech
Subject Code and Title : OML 751 Testing of Materials
Year and Branch : IV year Civil, EEE and ECE
Topics Covered : Unit -II: Mechanical Testing – Bend Test
Name of the Laboratory : Strength-of-Materials lab
Website Link : <http://sm-nitk.vlabs.ac.in/exp11/index.html>
Date : 11.09.2020
No.of Students participated : 61
Assignment given if any : -
(If Yes share the uploaded folder link)

Demonstration of Bend Test

OBJECTIVE

To study the behaviour of mild steel rod subjected to gradual increasing equal loads at 1/3rd span and to determine its mechanical properties.

Bending strength is defined as a material's ability to resist deformation under load, it represents the highest stress experienced within the material at its moment of rupture. There are two types of bending tests. Three-point bending test and four-point bending test. In a three-point bending test the area of uniform stress is quite small and concentrated under the centre loading point

The students can able to understand fundamental of Bending of Beam principle and applications on mild steel.

Course Outcome / Programme Outcomes	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO12
CO2 Analyze engineering components using various mechanical testing procedure.	3	1	1	1	2	2	2	1	2

STEPS:

1. When you click on a Bending test on mild steel, a new window will open as shown below.

BENDING TEST ON MILD STEEL ROD

Objective :
To study the behavior of the mild steel rod subjected to a gradual increasing equal loads at 1/3rd span and to determine mechanical properties.

Apparatus used:
Universal Testing Machine, Dial Indicators, Scale and Vernier calipers.

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Initial Diameter in two perpendicular directions

d_x (mm)	d_y (mm)	Average initial diameter d_i (mm)
$d_{1x}=26$	$d_{1y}=25.2$	25.6

Cross sectional area of the specimen $= A_i = \frac{\pi \times d_i^2}{4} = 514.7 \text{ mm}^2$

Moment of Inertia $= I = \frac{\pi \times d_i^4}{64} = [] \text{ mm}^4$

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Virtual Labs - Civil Engineering

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STRENGTH OF MATERIALS LAB

Home > Strength of Materials Lab > Bending Test on Mild Steel

Bending Test on Mild Steel

Test Your Knowledge

Question 1 of 5

1. As compared to uniaxial tension or compression, the strain energy stored in bending is only

Right answer

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